R13

Code No: 118EQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, May - 2019
SEMANTIC WEB AND SOCIAL NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	Define artificial intelligence.	[2]
b)	What are the objectives of semantic web architecture?	[3]
c)	List the famous ontology development tools.	[2]
d)	What is the purpose of visual modeling?	[3]
e)	Quote an example for Modus Ponens argument.	[2]
f)	An Expert system has three levels of organization. What are those levels?	[3]
g)	Give the four functions to be automated by OWL-S.	[2]
h)	What is the relationship between a service and profile?	[3]
i)	What is the drawback of links feature of web based network?	[2]
j)	What is the use of referral chaining in social network analysis?	[3]

PART - B

(50 Marks)

2. Discuss the development of information age and how the Web contributes. [10]

OR

3.a) Illustrate reasoning with semantic net.
b) What is meant by adaptive software? What is the need of soft computing for this? [10]

4. Explain the advantages of XML over HTML and also its limitations. How does RDF address these limitations? [10]

OR

5.a) What are the requirements for web ontology language? What is its relationship with RDF schema?
b) Differentiate between OWL-DL and OWL-Lite. [10]

6. "Non-monotonic rules are useful where information is unavailable." Substantiate this statement with illustrative examples. [10]

OR

7.a) Trade-off between finding the minimum necessary expressive power and the maximum possible reasoning capability for the semantic web.
b) What is the vision of educational semantic web? [10]

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8. Explain in detail the building blocks for the next generation of web services- Semantic web services. [10]

OR

9.a) What is social network? How is social network analysis a different approach to social phenomenon?

b) Discuss the role played by relational data in social network analysis. [10]

10.a) Is Jaccard coefficient a relative measure of co-occurrence? Justify your answer.

b) How to handle disambiguation of person names in social network mining? [10]

OR

11. Demonstrate a semantic web application development using Sesame, the RDF data repository. [10]

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